



CS\_316L

# WEB TECH LAB

## PROJECT FILE

### Shale Pizza - Online Food Ordering System

Technology Stack: HTML5, CSS3, Bootstrap 5, JavaScript, jQuery, PHP, MySQL

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January 2026

# 1. Introduction

## 1.1 Project Background

Shale Pizza is a local artisanal pizzeria aiming to expand its business through an online presence. The restaurant seeks to implement a modern, user-friendly web application that allows customers to browse the menu, customize orders, and place delivery/pickup orders online.

## 1.2 Problem Statement

- Current manual ordering process is time-consuming
- Limited customer reach beyond walk-in traffic
- Inefficient order management system
- No centralized customer database
- Difficulty in promoting special offers and tracking customer preferences

## 1.3 Project Objectives

1. Develop a responsive e-commerce platform for pizza ordering
2. Implement user authentication and profile management
3. Create an admin dashboard for order and inventory management
4. Integrate secure payment processing
5. Develop real-time order tracking system
6. Implement customer feedback and rating system

## 2. Features

### 2.1 Customer Features

- **User Registration & Login:** Secure authentication system
- **Menu Browsing:** Categorized pizza display with filtering options
- **Custom Pizza Builder:** Interactive tool for custom pizza creation
- **Shopping Cart:** Real-time cart updates with quantity management
- **Order Placement:** Delivery/pickup options with time scheduling
- **Order History:** Past orders with reorder functionality
- **Customer Reviews:** Rating and review system using forms
- **Loyalty Program:** Points-based reward system

### 2.2 Admin Features

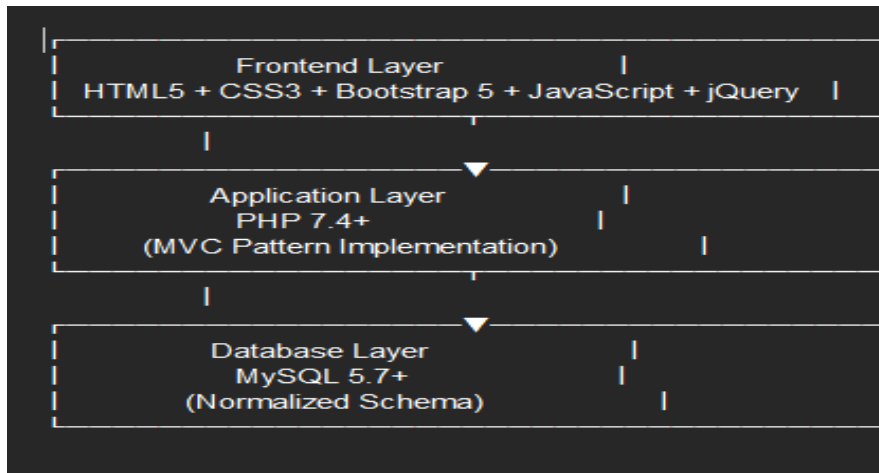
- **Dashboard:** Real-time sales analytics and metrics
- **Inventory Management:** Track ingredients and stock levels
- **Order Management:** Process, update, and cancel orders
- **Menu Management:** Add/edit/remove pizza items
- **Customer Management:** View customer data and order history
- **Promotion Management:** Create discount codes and special offers
- **Reports Generation:** Sales, customer, and inventory reports
- **Delivery Management:** Assign delivery personnel and track routes

### 2.3 System Features

- **Responsive Design:** Mobile and web approach
- **Search Functionality:** Quick pizza search with filters
- **Notifications:** Order confirmation, status updates
- **Social Media Integration:** Share orders and reviews

# 3. Architecture

## 3.1 System Architecture



## 3.2 Technical Architecture Components

Presentation Layer:

Responsive UI components

Client-side validation

Animations

**Business Logic Layer:**

PHP controllers

Service classes

Validation logic

Authentication middleware

**Data Access Layer:**

PDO database connections

Repository pattern

Query builders

**Database Layer:**

MySQL with InnoDB engine

Stored procedures for complex queries

Database backups and recovery

## 3.3 API Architecture

RESTful API design

JSON data format

Token-based authentication (JWT)

Rate limiting and throttling

API versioning

## 4. Workflow

### 4.1 Development Workflow

Phase 1: Requirements & Planning (Week 1-2)

- └─ Requirement gathering
- └─ Wireframing & prototyping
- └─ Database design
- └─ Technology stack finalization

Phase 2: Core Development (Week 3-8)

- └─ Setup development environment
- └─ Implement authentication system
- └─ Develop customer interface
- └─ Build admin dashboard
- └─ Integrate payment gateway

Phase 3: Testing & Deployment (Week 9-10)

- └─ Unit testing
- └─ Integration testing
- └─ Security testing

Phase 4: Maintenance (Ongoing)

- └─ Bug fixes
- └─ Feature enhancements
- └─ Performance optimization

### 4.2 User Registration Flow

Step 1: User visits registration page

Step 2: Fills registration form

Step 3: Client-side validation (JavaScript)

Step 4: Form submission to PHP

Step 5: Server-side validation

Step 6: Check for existing email/username

Step 7: Hash password

Step 8: Insert into users table

Step 9: Create session

Step 10: Redirect to homepage

### **4.3 Shopping Cart Flow**

- Step 1: User browses menu
- Step 2: Clicks "Add to Cart"
- Step 3: AJAX call to cart\_api.php
- Step 4: Check if cart exists, create if not
- Step 5: Add item to cart\_items table
- Step 6: Return success response
- Step 7: Update cart counter via jQuery
- Step 8: Show confirmation toast

### **4.4 Order Placement Flow**

- Step 1: User goes to checkout page
- Step 2: Review cart items and total
- Step 3: Enter delivery details
- Step 4: Submit order
- Step 5: Create order record in orders table
- Step 6: Transfer cart\_items to order\_items
- Step 7: Clear cart
- Step 8: Send confirmation email
- Step 9: Show order confirmation page

### **4.5 Admin Order Processing Flow**

- Step 1: Admin logs into dashboard
- Step 2: Views new orders (status: pending)
- Step 3: Clicks order to view details
- Step 4: Updates status to "confirmed"
- Step 5: Kitchen staff sees order
- Step 6: Updates status as order progresses
- Step 7: Marks as "delivered" when complete

# 5. Technology Implementation

## 5.1 Frontend (Client-side)

- HTML5: Semantic structure
- CSS3: Styling and animations
- Bootstrap 5: Responsive grid and components
- JavaScript: Interactive functionality
- jQuery: DOM manipulation and AJAX
- Font Awesome: Icons

## 5.2 Backend (Server-side)

- PHP 7.4+: Server-side scripting
- MySQL 5.7+: Database management
- Sessions: User state management

## 6.1 Development Environment

- **Local Setup:** XAMPP
- **Version Control:** Git with GitHub/GitLab
- **Code Editor:** VS Code with extensions
- **Local Testing:** PHP built-in server

### 1. Frontend Developer (2):

- HTML/CSS/Bootstrap implementation
- JavaScript/jQuery development
- Responsive design implementation

### 2. Backend Developer:

- PHP development
- Database design and implementation
- API development
- Security implementation